

LIBLAN[®] FILM COATED TABLET

Ingredient(s):

Each tablet contains:

Clidinium Bromide.....	2.5 mg
Chlordiazepoxide.....	5.0 mg

Pharmacodynamics:

1. Clidinium bromide is a quaternary ammonium antimuscarinic agent. It is an effective anticholinergic agent with activity approximating that of atropine sulfate against acetylcholine-induced spasms in isolated organs.
2. Chlordiazepoxide acts as an anxiolytic, sedative and hypnotic agent, depending on the dose. It also has a muscle relaxant action in cases of muscular spasticity and is an anticonvulsant.
3. The primary effect of chlordiazepoxide is to intensify the activity of the inhibitory transmitter γ -aminobutyric acid (GABA) in different parts of the central nervous system. Chlordiazepoxide causes no significant change in blood pressure, pulse rate, ECG or cardiac output.

Pharmacokinetics:

Chlordiazepoxide is completely absorbed after oral administration usually within 1 and 2 hours. The systemic bioavailability of an oral dose is close to 100%. Chlordiazepoxide is extensively bound to plasma proteins and the elimination half-life is about 5 to 30 hours. Protein binding is reduced in liver disease and chronic renal failure. Chlordiazepoxide is metabolized by oxidation to active and inactive metabolites before final inactivation as glucuronide conjugates. It is excreted via urine and faeces.

Indication(s):

Adjunctive therapy in the treatment of peptic ulcer and irritable bowel syndrome; hypersecretion and hypermotility of the gastrointestinal tract.

Dosage and Administration:

Oral: 1 to 2 tablets one to four times daily, 30 to 60 minutes before meal, dose being adjusted as needed and tolerated. Usual adult prescribing limit is 8 tablets a day.

Elderly and debilitated patients, initially 1-2 tablets daily, the dosage then being adjusted to most effective well-tolerated dose.

Missed dose: take as soon as possible but do not take if almost time for next dose, not doubling doses.

To be dispensed on physician's prescription. Avoid alcoholic beverages.

Contraindication(s):

Hypersensitivity to chlordiazepoxide and / or clidinium bromide, glaucoma, prostatic hypertrophy, benign bladder neck obstruction, myasthenia gravis, obstructive gastrointestinal disease, paralytic ileus or intestinal atony, ulcerative colitis or toxic megacolon and first trimester pregnancy.

Precaution(s) / Warning(s):

1. Caution patients about possible combined effects with alcohol and other CNS depressants, particularly patients involved in occupations that require complete mental alertness (e.g. operating machinery, driving).
2. In the elderly and debilitated, limit dose to smallest effective amount to preclude ataxia, oversedation and confusion. If combination therapy with other psychotropics seems indicated, carefully consider pharmacology of these agents, particularly potentiating agents such as MAOI, phenothiazines.
3. Physical and psychological dependence have been reported at recommended doses of benzodiazepines, caution should be exercised in prescribing for individuals known to be addiction prone.
4. Use with caution in individuals with evidence of impending depression and/or suicidal tendencies as chlordiazepoxide in this preparation essentially acts on the anxiety component of depression and should not be used as alone for treating depression.
5. Caution should be exercised in prescribing for patients with pre-existing autonomic neuropathy as the drug may aggravate the pre-existing disease.
6. Caution should be exercised in prescribing for patients with cardiovascular disorders such as congestive heart failure, coronary heart disease, hypertension, tachycardia, and tachyarrhythmias. In these cases, the dosage may need to be adjusted according to patient's response.
7. Use with caution in patients with hyperthyroidism.
8. Use with caution in the presence of impaired renal or hepatic function.
9. Heat prostration may occur with the use of anticholinergic due to decreased sweating.
10. Paradoxical reactions to chlordiazepoxide have been reported in psychiatric patients and should be administered with caution.
11. Elderly may exhibit increased susceptibility anticholinergic effect of clidinium and to CNS effect of chlordiazepoxide.
12. **Risks from Concomitant Use with Opioids**
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of chlordiazepoxide with opioids. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these

risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when clordiazepoxide is used with opioids. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the opioid have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of opioids (See Drug Interactions).

Caution – This preparation may be habit forming on prolonged use.

Use in Pregnancy and Lactation:

Safe use of this product in pregnancy has not been established. It should not be used in the first or last trimester without compelling reasons.

Chlordiazepoxide is excreted in breast milk, clidinium may cause inhibition of lactation.

Use in Children:

Children may exhibit increased susceptibility to anticholinergic effects of clidinium and to CNS effects of chlordiazepoxide. Its use in children is generally not recommended unless the potential benefit outweighs the potential risk.

Side Effect(s) / Adverse Reaction(s):

Drowsiness, ataxia and confusion have been observed in some patients, particularly the elderly and debilitated. These effects can be avoided in almost all instances by proper dosage adjustment.

Other adverse reactions have been infrequent including skin eruptions, edema, minor menstrual irregularities, nausea and constipation, dry mouth, extrapyramidal symptoms, agranulocytosis, granulocytopenia, or leukopenia, allergic reaction, CNS depression, increased intraocular pressure, jaundice, blurred vision and urinary retention. Such effects are generally controlled with reduction of dosage.

Drug Interaction(s):

1. Concurrent use with anticholinergic may intensify anticholinergic effects. Patients should be advised to report occurrence of gastrointestinal problem promptly since paralytic ileus may occur with concurrent therapy.
2. Clidinium bromide may reduce absorption of ketoconazole and itraconazole. Patients should be advised to take medication at least 2 hours after administration of ketoconazole.
3. Antagonize effect of metoclopramide and cisapride on gastrointestinal motility if used concurrently.
4. Opioid – concurrent use with clidinium bromide may result in high risk of severe constipation, which subsequently may lead to paralytic or urinary retention.
5. Sedative effect may be enhanced when chlordiazepoxide is used concurrently with lofexidine, alcohol, antipsychotic, antihistamine and anaesthetic. It may also enhance hypotensive effect of antihypertensives; and sedative effect of chlordiazepoxide may be enhanced when used with alpha-blockers, possibly moxonidine.

6. Opioids

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABA_A sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Precautions / Warnings).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Symptoms and Treatment for Overdosage and Antidote(s):

Symptoms of overdose indicating need for medical attention include continuing confusion, decreased reflexes, severe drowsiness, slow heartbeat, shortness of breath or troubled breathing, continuing slurred speech, staggering and severe weakness. There is no specific antidote for its overdose, therefore, management of the patient

consist of symptomatic and supportive therapy. The anticholinergic side effects of clidinium bromide may be treated with pilocarpine while hypotension may be treated with noradrenaline.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Keep in a tight container. Store at temperature below 30°C. Protect from light and moisture.

Product Description & Packing(s):

Light green colored film coated tablet.

Plastic bottle of 1000's (for export only).

Blister packing of 10's x 100.



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